



PHYSICAL DATA MODEL (PDM)

Document 04 of 09 · B2B Data Engineering Platform



Specifies how the model is implemented in PostgreSQL: schemas, tables, data types, keys, constraints, indexes and partitions.

Document ID	B2B-DEP-04
Version	1.3
Status	Final
Date	05/10/2026
Prepared by	Syed Kazim Raza
Reviewed by	[Team Lead]

05/10/2026

DATAROPES.AI

Data Engineering Internship · B2B Data Engineering Platform

Document Control

Revision History

Version	Date	Author	Description of Change
0.1	-	Syed Kazim Raza	Template created
1.0	28/09/2026	Syed Kazim Raza	Content populated from DDL scripts and the PostgreSQL catalog (constraints, indexes, partitions)
1.1	30/09/2026	Syed Kazim Raza	Row counts of the rebuilt data; staging emptied after each run; rejected_records; DEFAULT partition; pk_fact_leads
1.2	04/10/2026	Syed Kazim Raza	PDM diagram redrawn (light theme, draw.io); design notes show dimension CLUSTER and DEFAULT partition
1.3	05/10/2026	Syed Kazim Raza	Formatting to documentation standard: table column widths, numbered figure and table captions with lists, figure descriptions, complete acronym list

Approvals

Role	Name	Signature	Date
Prepared by	Syed Kazim Raza		
Reviewed by	[Team Lead]		
Approved by	[Manager]		

Related Documents

Ref	Document	Location
R1	Business Requirements Document (BRD)	Provided by DataRopes
R2	Project Document (technical reference)	docs/deliverables/project_document.md
R3	Project Plan	docs/deliverables/project_plan.md
R4	README / Deployment Guide	README.md

Contents

Document Control 2

Contents 3

List of Figures..... 4

List of Tables..... 5

1. Introduction 6

 1.1 Purpose 6

 1.2 Scope 6

 1.3 Intended Audience 6

 1.4 Definitions & Acronyms 6

2. Platform & Environment 7

3. Layered Schema Design 7

4. Physical Data Model Diagram..... 8

5. Table Inventory..... 9

6. Column Specifications 9

7. Keys & Constraints..... 11

8. Indexing Strategy 12

9. Partitioning Strategy..... 13

10. Surrogate Keys & SCD Implementation 14

11. Naming Conventions 17

12. DDL Deployment Order 17

13. Assumptions, Constraints & Open Issues 17

List of Figures

Figure 1: Physical data model — warehouse schema 8

Figure 2: Monthly partitions of fact_orders, plus the DEFAULT partition..... 14

Figure 3: dim_companies — SCD Type 2 versions of the same company..... 16

List of Tables

Table 1: Intended Audience..... 6

Table 2: Definitions & Acronyms 6

Table 3: Platform & Environment..... 7

Table 4: Layered Schema Design 7

Table 5: Table Inventory 9

Table 6: Column Specifications..... 9

Table 7: Keys & Constraints 11

Table 8: Indexing Strategy 12

Table 9: Partitioning Strategy 13

Table 10: Naming Conventions..... 17

Table 11: DDL Deployment Order 17

Table 12: Assumptions, Constraints & Open Issues 17

1. Introduction

1.1 Purpose

This document describes the physical implementation of the data model across all PostgreSQL layers.

1.2 Scope

In scope:

- Staging, intermediate, warehouse, marts and metadata schemas
- Tables, columns, data types, keys, constraints, indexes, partitions, naming

Out of scope:

- Business definitions (see LDM)
- Source-to-target transformation rules (see SMX)

1.3 Intended Audience

Table 1: Intended Audience

Audience	How they use this document
Data Engineers / DBAs	Create, maintain and tune the database
Reviewers	Check design against standards

1.4 Definitions & Acronyms

Table 2: Definitions & Acronyms

Term	Definition
B2B	Business-to-Business: companies selling to other companies.
BI	Business Intelligence: reporting and dashboards, here Power BI.
BRD	Business Requirements Document: here the Project Requirements Document provided by DataRopes.
CSV	Comma-Separated Values: the text-file format of the marketing leads and web logs feeds.
DDL	Data Definition Language: SQL that creates schemas, tables, indexes and views.
Degenerate dimension	A business key kept on a fact without its own dimension table (order_id, lead_id).
DEP	Data Engineering Platform: prefix of the document IDs (B2B-DEP-01 ...).
FK	Foreign Key: a column that references the primary key of another table.
HTTP	Hypertext Transfer Protocol: the web protocol recorded in the web logs (GET, POST ...).
ID	Identifier.
KPI	Key Performance Indicator: a business measure shown on the dashboard (KPI-01 ... KPI-15).
LDM	Logical Data Model (deliverable 03).
MERGE	PostgreSQL 15+ statement that inserts or updates in one pass; used by all loads.
Partial index	An index that covers only rows matching a WHERE condition (e.g. is_current = TRUE).
PDM	Physical data model: tables, columns, data types, keys, indexes and partitions as implemented in PostgreSQL.
PK	Primary Key: the column(s) that uniquely identify a row.
Range partitioning	Splitting a table into child tables by value ranges of a column (here: one partition per month of date_key).

Term	Definition
SCD	Slowly Changing Dimension: SCD1 overwrites changes, SCD2 keeps every version with validity dates.
SERIAL	PostgreSQL auto-incrementing integer column, used for surrogate keys.
SMX	Source-to-Target Mapping workbook (deliverable 05, Excel).
SQL	Structured Query Language.
UTF-8	Unicode character encoding used by the CSV files.
UUID	Universally Unique Identifier: the 32-character IDs used as business keys.
YYYYMMDD	Date written as year, month, day (e.g. 20260930); the format of date_key.

2. Platform & Environment

Table 3: Platform & Environment

DBMS	PostgreSQL
Version	18.4 (MERGE requires 15+)
Database	b2b_warehouse_db (name from .env POSTGRES_DATABASE)
Schemas	staging, intermediate, warehouse, marts, metadata
Encoding	UTF8
Tooling	pgAdmin 4 / psql; Python SQLAlchemy + psycopg2 for loads; Power BI Desktop for reporting

3. Layered Schema Design

Table 4: Layered Schema Design

Schema	Purpose	Constraints	Load Pattern	Retention
staging	Raw landing copy of the sources	None (no PK/FK, all nullable); _loaded_at NOT NULL DEFAULT now()	Append (SQL Server) / TRUNCATE + append (CSV); emptied after each run	Latest batch only (48 kB after a run)
intermediate	Clean, validated, latest version per key	PK, FK, UNIQUE and CHECK constraints	MERGE upsert (idempotent)	Current state only
warehouse	Dimensional model for reporting	PK, FK, UNIQUE, CHECK; partial unique index for SCD2	MERGE (SCD1 / SCD2 / facts)	Full history; partitions 2 years back to 6 months ahead
marts	15 KPI views for Power BI (+2 pipeline views)	Views only (no storage)	CREATE OR REPLACE VIEW	n/a
metadata	Watermarks and run log	PK on pipeline_name / run_id	Insert / update by pipelines	Unlimited

4. Physical Data Model Diagram

Figure 1 shows the physical warehouse schema with data types, primary and foreign keys and indexed columns.

Physical Data Model (PDM) — PostgreSQL warehouse schema

Galaxy schema: 4 fact tables (range-partitioned monthly on date_key) · 5 dimensions · 1 bridge · data types, keys and indexes as deployed

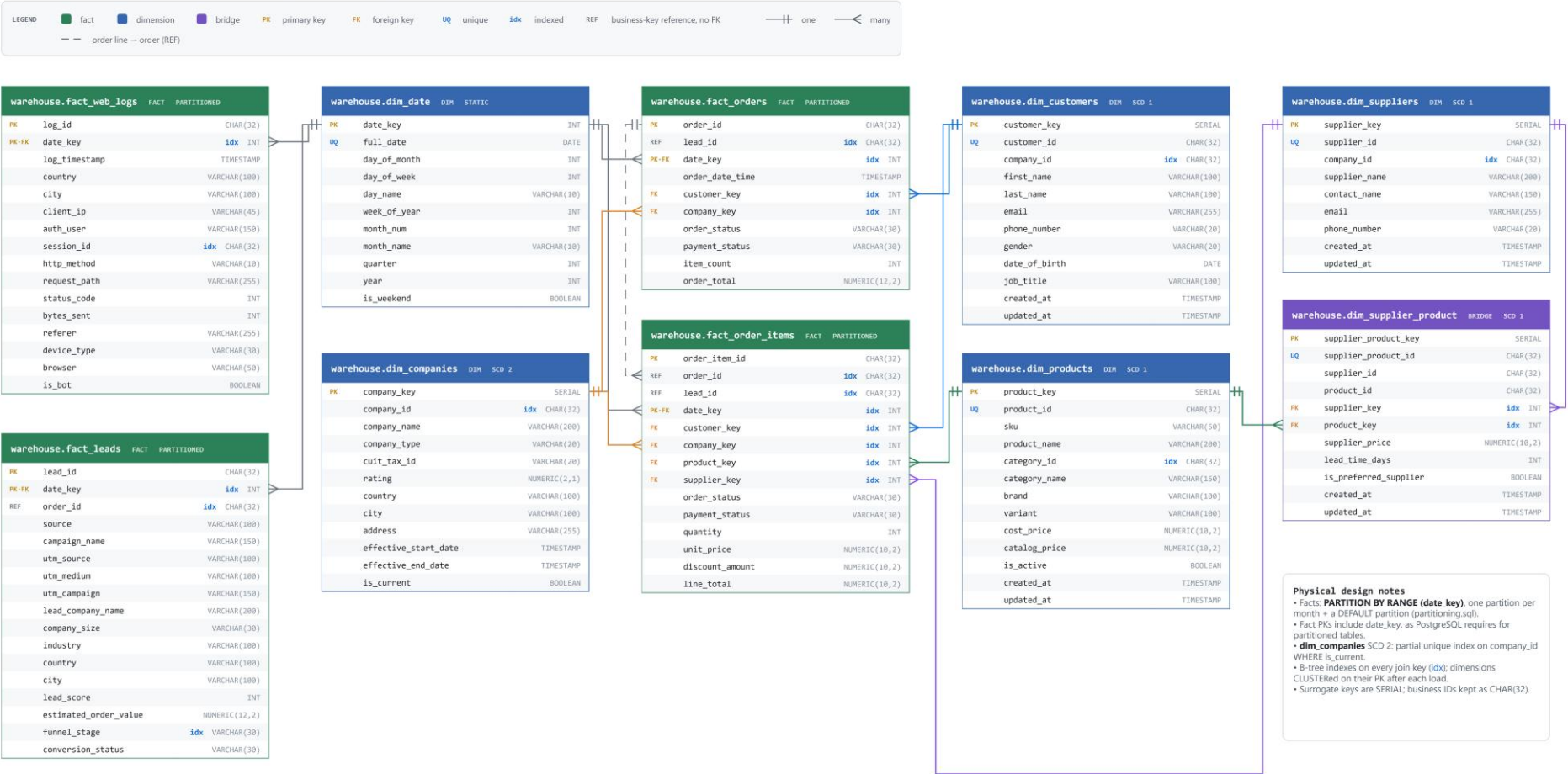


Figure 1: Physical data model — warehouse schema

5. Table Inventory

Table 5: Table Inventory

Schema	Table	Type	Grain	Primary Key	Partitioning	Est. Rows	DDL
warehouse	dim_date	Dimension (generated)	One row per calendar day	date_key	-	912	\$12/6
warehouse	dim_companies	Dimension - SCD2	One row per company version	company_key	-	2,505	\$12/6
warehouse	dim_customers	Dimension - SCD1	One row per customer	customer_key	-	94,755	\$12/6
warehouse	dim_suppliers	Dimension - SCD1	One row per supplier	supplier_key	-	400	\$12/6
warehouse	dim_products	Dimension - SCD1	One row per product	product_key	-	96	\$12/6
warehouse	dim_supplier_product	Dimension / bridge - SCD1	One row per supplier-product offer	supplier_product_key	-	347	\$12/6
warehouse	fact_orders	Fact - transaction	One row per order	order_id, date_key	Monthly x31 + default	156,678	\$12/6
warehouse	fact_order_items	Fact - transaction	One row per order line	order_item_id, date_key	Monthly x31 + default	470,263	\$12/6
warehouse	fact_web_logs	Fact - event	One row per HTTP request	log_id, date_key	Monthly x31 + default	296,703	\$12/6
warehouse	fact_leads	Fact - accumulating	One row per lead	lead_id, date_key	Monthly x31 + default	148,358	\$12/6
metadata	pipeline_watermarks / pipeline_run_log	Control / audit	One row per watermark key / per run	pipeline_name / run_id	-	8 / 20	\$12/3
staging	10 tables (companies ... web_logs)	Landing	Latest batch (emptied after each run)	None	-	0 after a run	\$12/4
intermediate	10 tables (same names) + rejected_records	Clean	One row per business key	Business key	-	1,170,110 (+ 6,775 rejected)	\$12/5

6. Column Specifications

The full column-level specification of all 43 tables (458 columns: type, nullability, default, keys, description, example, SMX mapping ID) is maintained in **04b_Data_Dictionary.xlsx**. The two most important tables are summarised below: the SCD2 company dimension and the order fact.

Table 6: Column Specifications

Table	Column	Data Type	Null	Default	Key	Description
dim_companies	company_key	SERIAL	N		PK	Surrogate key of the company version (SCD2).
dim_companies	company_id	CHAR(32)	N			Business identifier of the company (32-char hex UUID assigned by the Python generator).
dim_companies	company_name	VARCHAR(200)	N			Legal/trading name of the company.

Table	Column	Data Type	Null	Default	Key	Description
dim_companies	company_type	VARCHAR(20)	N			Role of the company in the marketplace: Buyer or Supplier.
dim_companies	cuit_tax_id	VARCHAR(20)	N			Company tax identification number (unique per company).
dim_companies	Rating	NUMERIC(2,1)	N			Company rating on a 1.0-5.0 scale.
dim_companies	Country	VARCHAR(100)	N			Country name.
dim_companies	City	VARCHAR(100)	N			City name.
dim_companies	Address	VARCHAR(255)	N			Street address of the company.
dim_companies	effective_start_date	TIMESTAMP	N			Start of the validity period of this company version.
dim_companies	effective_end_date	TIMESTAMP	Y			End of the validity period (NULL = current version).
dim_companies	is_current	BOOLEAN	N	TRUE		TRUE for the current company version.
fact_orders	order_id	CHAR(32)	N		PK	Business identifier of the order.
fact_orders	lead_id	CHAR(32)	Y			Identifier of the marketing lead that generated the record (NULL if none).
fact_orders	date_key	INT	N		PK/FK	Date surrogate key in YYYYMMDD format (FK to dim_date); also the partition key on facts.
fact_orders	order_date_time	TIMESTAMP	N			Date and time the order was placed (renamed from order_date).
fact_orders	customer_key	INT	N		FK	Surrogate key of the customer.
fact_orders	company_key	INT	N		FK	Surrogate key of the company version (SCD2).

Table	Column	Data Type	Null	Default	Key	Description
fact_orders	order_status	VARCHAR(30)	N			Fulfilment status of the order.
fact_orders	payment_status	VARCHAR(30)	N			Payment status of the order.
fact_orders	item_count	INT	N			Number of order lines in the order.
fact_orders	order_total	NUMERIC(12,2)	N			Total order amount (sum of line totals).

7. Keys & Constraints

Table 7: Keys & Constraints

Constraint	Table	Type	Definition
<table>_pkey	All dimensions	PRIMARY KEY	Surrogate key (company_key, customer_key, ...); dim_date on date_key
pk_fact_orders / pk_fact_order_items / pk_fact_web_logs / pk_fact_leads	Facts	PRIMARY KEY	(business key, date_key) - partition key must be part of the PK
<table>_<business_key>_key	dim_customers, dim_suppliers, dim_products, dim_supplier_product, dim_date	UNIQUE	Business key unique (customer_id, supplier_id, product_id, supplier_product_id, full_date)
uq_dim_companies_current	dim_companies	UNIQUE partial index	UNIQUE (company_id) WHERE is_current = TRUE - one current version per company
uq_foi_order_supplier_product	fact_order_items	UNIQUE	(order_id, supplier_key, product_key, date_key) - no duplicate product per order
fk_fo_* / fk_foi_* / fk_fwl_date / fk_fl_date	Facts	FOREIGN KEY	date_key -> dim_date; customer_key, company_key, product_key, supplier_key -> dimensions (14 FKs)
fk_dsp_supplier / fk_dsp_product	dim_supplier_product	FOREIGN KEY	supplier_key -> dim_suppliers, product_key -> dim_products

Constraint	Table	Type	Definition
CHECK constraints	Dimensions and facts	CHECK	Domains (company_type, gender, order/payment status, conversion_status), ranges (rating, lead_score, status_code, calendar parts), non-negative amounts, catalog_price >= cost_price, SCD2 end > start

8. Indexing Strategy

Table 8: Indexing Strategy

Index	Table	Column(s)	Type	Purpose
uq_dim_companies_current	dim_companies	company_id WHERE is_current	Unique partial B-tree	Enforce one current version; fast current lookup
idx_dim_companies_id	dim_companies	company_id	B-tree	Point-in-time lookup of company versions in fact loads
idx_dim_customers_company	dim_customers	company_id	B-tree	Customer-to-company joins
idx_dim_suppliers_company	dim_suppliers	company_id	B-tree	Supplier-to-company joins
idx_dim_products_category	dim_products	category_id	B-tree	Filter / group by category
<table>_pkey (CLUSTER)	All 6 dimensions	Surrogate key	B-tree + CLUSTER	Rows stored in key order after each load (run_pipeline.py), as the requirements ask
idx_dsp_supplier, idx_dsp_product	dim_supplier_product	supplier_key; product_key	B-tree	FK joins
idx_fo_date / customer / company / lead	fact_orders	date_key; customer_key; company_key; lead_id	B-tree (per partition)	Date filters, FK joins, lead-to-order link
idx_foi_date / customer / company / product / supplier / lead / order	fact_order_items	date_key; customer_key; company_key; product_key; supplier_key; lead_id; order_id	B-tree (per partition)	Date filters, FK joins, order roll-up
idx_fwl_date, idx_fwl_session	fact_web_logs	date_key; session_id	B-tree (per partition)	Date filters; session lookups
idx_fl_date / order / funnel	fact_leads	date_key; order_id; funnel_stage	B-tree (per partition)	Date filters, conversion join, funnel grouping

9. Partitioning Strategy

Table 9: Partitioning Strategy

Table	Partition Key	Scheme	Granularity	Maintenance
fact_orders	date_key (INT YYYYMMDD)	RANGE, FROM (YYYYMM01) TO (next month)	Monthly - 31 partitions (2024_09 ... 2027_03) + 1 DEFAULT	partitioning.sql runs on every warehouse load: CREATE TABLE IF NOT EXISTS for 2 years back to 6 months ahead; no automatic drop
fact_order_items	date_key (INT YYYYMMDD)	RANGE, FROM (YYYYMM01) TO (next month)	Monthly - 31 partitions (2024_09 ... 2027_03) + 1 DEFAULT	partitioning.sql runs on every warehouse load: CREATE TABLE IF NOT EXISTS for 2 years back to 6 months ahead; no automatic drop
fact_web_logs	date_key (INT YYYYMMDD)	RANGE, FROM (YYYYMM01) TO (next month)	Monthly - 31 partitions (2024_09 ... 2027_03) + 1 DEFAULT	partitioning.sql runs on every warehouse load: CREATE TABLE IF NOT EXISTS for 2 years back to 6 months ahead; no automatic drop
fact_leads	date_key (INT YYYYMMDD)	RANGE, FROM (YYYYMM01) TO (next month)	Monthly - 31 partitions (2024_09 ... 2027_03) + 1 DEFAULT	partitioning.sql runs on every warehouse load: CREATE TABLE IF NOT EXISTS for 2 years back to 6 months ahead; no automatic drop

Figure 2 lists the monthly partitions of fact_orders and the DEFAULT partition that catches unexpected dates.

```

60  SELECT
61      inhrelid::regclass AS partition
62  FROM pg_inherits
63  WHERE inhparent = 'warehouse.fact_orders'::regclass
64  ORDER BY 1;
65  --(31 months + fact_orders_default)
66

```

Data Output Messages Notifications

	partition regclass	
15	warehouse.fact_orders_2025_11	
16	warehouse.fact_orders_2025_12	
17	warehouse.fact_orders_2026_01	
18	warehouse.fact_orders_2026_02	
19	warehouse.fact_orders_2026_03	
20	warehouse.fact_orders_2026_04	
21	warehouse.fact_orders_2026_05	
22	warehouse.fact_orders_2026_06	
23	warehouse.fact_orders_2026_07	
24	warehouse.fact_orders_2026_08	
25	warehouse.fact_orders_2026_09	
26	warehouse.fact_orders_2026_10	
27	warehouse.fact_orders_2026_11	
28	warehouse.fact_orders_2026_12	
29	warehouse.fact_orders_2027_01	
30	warehouse.fact_orders_2027_02	
31	warehouse.fact_orders_2027_03	
32	warehouse.fact_orders_default	

Total rows: 32 Query complete 00:00:00 137

Figure 2: Monthly partitions of fact_orders, plus the DEFAULT partition

10. Surrogate Keys & SCD Implementation

- **Surrogate keys:** every dimension except dim_date has a SERIAL integer key (company_key, customer_key, supplier_key, product_key, supplier_product_key). Facts store only these integer keys,

which keeps fact rows narrow and joins fast. The business key (32-char id) is kept in the dimension with a UNIQUE constraint (SCD1 dimensions).

- **dim_date** uses a smart integer key YYYYMMDD, which doubles as the fact partition key and makes partition ranges readable.
- **SCD Type 1** (dim_customers, dim_suppliers, dim_products, dim_supplier_product): `MERGE ... WHEN MATCHED AND (a IS DISTINCT FROM b OR ...) THEN UPDATE` - the row is overwritten only when an attribute actually changed; IS DISTINCT FROM makes the comparison NULL-safe.
- **SCD Type 2** (dim_companies, 2,505 rows for 2,500 companies): step 1 MERGE expires the current row (`effective_end_date = source.updated_at, is_current = FALSE`) when one of 7 tracked attributes changed; step 2 inserts a new current row for every company without a current row. The first version starts at `created_at`, later versions at `updated_at`.
- **Integrity of SCD2** is enforced physically by the partial unique index `uq_dim_companies_current` (one current row per `company_id`) and the CHECK `effective_end_date IS NULL OR effective_end_date > effective_start_date`.
- **As-of join in fact loads:** `fact_orders` and `fact_order_items` pick the company version valid at order time:

```
JOIN warehouse.dim_companies com
  ON com.company_id = o.company_id
 AND o.order_date >= com.effective_start_date
 AND (com.effective_end_date IS NULL OR o.order_date < com.effective_end_date)
```


11. Naming Conventions

Table 10: Naming Conventions

Object	Convention	Example
Schema	Lower-case layer name	staging, intermediate, warehouse, marts, metadata
Staging / intermediate table	Lower-case plural source entity name	orders, order_items, supplier_product_mapping
Dimension / fact table	dim_<entity plural> / fact_<process>	dim_companies, fact_order_items
Surrogate / business key	<entity>_key (INT) / <entity>_id (CHAR(32))	company_key / company_id
Date key	date_key INT YYYYMMDD	20260928
Partition	<fact>_YYYY_MM	fact_orders_2026_09
Constraint / index	pk_<table>, fk_<abbr>_<ref>, uq_<...>, idx_<abbr>_<column>	pk_fact_orders, fk_foi_product, idx_fo_date
Mart view	marts.vw_<subject>	marts.vw_monthly_revenue_trend
Audit column	_loaded_at (leading underscore = technical)	staging.orders._loaded_at

12. DDL Deployment Order

Table 11: DDL Deployment Order

Order	Script	Creates
1	sql_server/source1/ddl-source1.sql	b2b_source_db and 8 source tables (drops and recreates the database)
2	sql_server/metadata/ddl-metadata.sql	metadata.Generation_Seeds (SQL Server)
3	postgresql/metadata/metadata_ddl.sql	pipeline_watermarks, pipeline_run_log
4	postgresql/staging/staging_ddl.sql	10 staging tables
5	postgresql/intermediate/DDL/intermediate_ddl.sql	10 intermediate tables + rejected_records, with constraints
6	postgresql/warehouse/DDL/warehouse_ddl.sql	6 dimensions, 4 partitioned facts, 23 indexes
7	postgresql/warehouse/PARTITIONS_SETUP/partitioning.sql	Monthly fact partitions + DEFAULT (re-run on every load)
8	postgresql/marts/marts_ddl.sql	15 KPI views (17 views in marts)

13. Assumptions, Constraints & Open Issues

Table 12: Assumptions, Constraints & Open Issues

ID	Type	Description	Owner	Status
A1	Assumption	Facts fall inside the partition window (2 years back to 6 months ahead); any other date goes to the DEFAULT partition, which stays empty.	Data Engineering	Accepted
A2	Assumption	Staging is emptied after each successful run; the sources keep all data if a reload is needed.	Data Engineering	Accepted